

ReadMe

Empirical Replication Project: Forecasting the Term Structure of South African Government Bond Yields

Replication of Diebold, F.X. and Li, C. (2006), "Forecasting the term structure of government bond yields", Journal of Econometrics 130, 337–364.

DDM8X01 Debt Market Modelling, Semester Test 2 (Empirical Component). Group 11: Bukho Ndziweni (220081696) and Sizwe November (200819365).

1. Overview

The analysis estimates the Nelson-Siegel three-factor model (level, slope, curvature) on monthly South African government bond and Treasury bill yields, models each factor as an AR(1) (first-order autoregressive) process, and evaluates recursive out-of-sample forecasts at 1, 6 and 12-month horizons against benchmark models. The main sample is January 1994 to February 2023; robustness checks extend to May 2026. All results in the report are produced by the code in the Code folder from the raw data in the Data/Raw folder.

2. Software and packages

Analysis run with Python in Jupyter Notebook (Anaconda distribution) on Windows. Versions used:

Software / package	Version and purpose
Python	3.13.9
pandas	2.3.3 (data handling)
numpy	2.3.5 (numerical calculations, OLS)
statsmodels	0.14.5 (autocorrelations, ADF tests, AR(1) regressions, Newey-West standard errors)
matplotlib	3.10.6 (figures)
scipy	1.16.3 (normal distribution p-values for Diebold-Mariano tests)
openpyxl	Any recent version (reading the SARB Excel file; included with Anaconda)
requests	Any recent version (data download notebook only; included with Anaconda)

All packages are included in a standard Anaconda installation. If any are missing, install them with:

```
pip install pandas numpy statsmodels matplotlib scipy openpyxl requests
```

Figures use the Calibri font. On a computer without Calibri, matplotlib substitutes a default font and prints a warning; no results are affected.

3. Folder structure

Keep the folder structure unchanged. The code uses relative paths (for example ../Data/Raw/) and must be run from inside the Code folder.

File	Description
Data/Raw/SARB_govt_bond_yields_monthly.xlsx	Raw SARB monthly government bond yields (KBP2000M, KBP2001M, KBP2002M, KBP2003M, KBP2049M), exactly as

File	Description
	downloaded
Data/Raw/SARB_treasury_bills_daily.csv	Raw SARB daily Treasury bill tender rates (MMRD203A, MMRD206A, MMRD209A, MMRD212A), saved unchanged by 00_download_tsbills.ipynb
Data/Raw/Data_Sources.txt	Sources, series codes, descriptions, units, date ranges and download date
Code/00_download_tsbills.ipynb	Downloads the Treasury bill rates from the SARB web data service
Code/01_replication.ipynb	Main analysis: data preparation, estimation, forecasting, diagnostics and robustness (Steps 1 to 11)
Code/01_replication.py	Plain Python script version of 01_replication.ipynb
Output/01_replication_output.html (.pdf)	Complete, unedited software output of the main analysis
Output/00_download_tsbills_output.html	Complete, unedited software output of the download notebook
Output/sa_yields_monthly_clean.csv	Cleaned monthly yield panel, all maturities, January 1994 to May 2026
Output/ns_factors_main.csv	Estimated Nelson-Siegel factors, main specification
Output/ns_factors_all_maturities.csv	Estimated factors using all available maturities (robustness)
Output/fig1 to fig8 (.png)	Figures used in the report
Report/	Empirical report (PDF)

4. Execution order

1. **00_download_tsbills.ipynb** (optional). Downloads the Treasury bill rates. This step does not need to be repeated: the downloaded file is already in Data/Raw. SARB revises and extends its data, so a new download may differ slightly from the file used in the report.
2. **01_replication.ipynb** (required). Reads the two raw files and produces every table, figure and statistic in the report. Runtime is under one minute.

5. Instructions to reproduce the analysis

Option A: Jupyter Notebook (recommended)

1. Install Anaconda (or Python with the packages in Section 2).
2. Open Jupyter Notebook and navigate to the Code folder of the project.
3. Open 01_replication.ipynb.
4. Select Kernel, then Restart & Run All. The notebook runs from start to finish without manual intervention.

Option B: Python script

1. Open Anaconda Prompt (or a terminal) and change directory to the Code folder.
2. Prevent figure windows from pausing the script: on Windows run `set MPLBACKEND=Agg` (on Mac or Linux: `export MPLBACKEND=Agg`).

3. Run `python 01_replication.py`. Tables print to the console and figures are saved to the Output folder.

6. Notes on data preparation

- Raw files are never edited. All cleaning is done in code: skipping SARB metadata rows, treating “undefined” as missing, converting dates to month-ends, and averaging daily Treasury bill rates to monthly.
- Treasury bill discount rates are converted to yields using $y = d / (1 - d \times n/365)$.
- Maturity bands are assigned band-midpoint maturities (18, 48, 90 and 300 months); “10 years and over” is assigned 150 months, with 180 months tested in Robustness A.
- The SARB 3 to 5 year series (KBP2001M) records zeros from March 2023 to January 2025. These are set to missing in code, and the main specification ends in February 2023. Robustness B uses all available maturities to May 2026.
- Tables printed by the code are numbered to follow Diebold and Li (2006): Tables 1 to 7, plus Table 8 (diagnostic Diebold-Mariano tests).